Southeast Louisiana Pipeline Infrastructure Analysis

Project MASABB Site Assessment

SESCO Cement Internal Document

Prepared For: Port Nickel Development Team

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Site Location: Braithwaite, Plaquemines Parish, Louisiana

387 Acres, Mississippi River Mile 80 (East Bank)

EXECUTIVE SUMMARY

Purpose

This report analyzes oil and gas pipeline infrastructure in Southeast Louisiana to evaluate connectivity opportunities for the Port Nickel site (387 acres, Braithwaite, Plaquemines Parish). The analysis identifies existing and planned pipeline systems within the region, assesses proximity to the site, and evaluates strategic positioning within Louisiana's expanding energy infrastructure corridor.

Key Findings

Regional Context:

- Southeast Louisiana hosts 12+ major natural gas transmission pipelines driven by LNG export growth (15.3 Bcf/d operational, expanding to 28+ Bcf/d by 2028)
- St. James Parish (60 miles upriver) serves as Louisiana's primary crude oil terminal hub
- All recent pipeline construction (2023-2027) is LNG-driven; 8 major natural gas projects under development

Site Proximity:

- **Gulf South Pipeline** (major intrastate gas system): 10-25 miles estimated
- **Harvest CAM System**: <5 miles - serves Belle Chasse crude oil terminal (Harvest Midstream)
- **Transco, Colonial, Collins**: 15-30 miles (East Bank corridor)
- **Plaquemines LNG feed pipelines**: 50-75 miles south (same parish)

Strategic Position:

- East Bank location provides direct access to major pipeline corridor without river crossings
- Positioned between New Orleans/Baton Rouge refining complex (north) and Plaquemines LNG terminals (south)
- Same parish as Plaquemines LNG (Venture Global) - largest LNG growth driver in region

Plaquemines LNG Feed Infrastructure:

- **Gator Express Pipeline** (~2.0 Bcf/d header pipeline) serves Plaquemines LNG in same parish
- **East Lateral Xpress** (0.725 Bcf/d, Columbia Gulf/TC Energy) operational 2025, ties into Gator Express
- Plaquemines receives feedgas from three sources: TETCO, Tennessee Gas Pipeline, Columbia Gulf

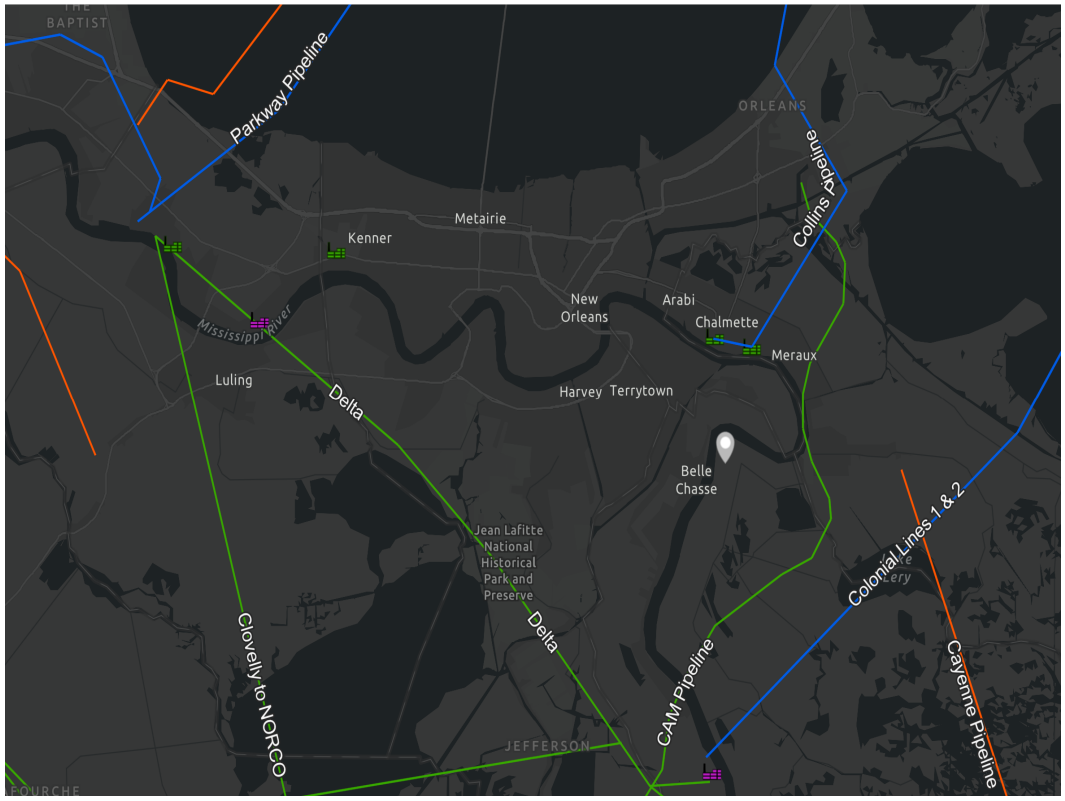

Figure: Port Nickel Site - Pipeline Infrastructure (Belle Chasse Area)

Strategic Implications

Port Nickel's East Bank location positions the site along established pipeline corridors serving Louisiana's LNG export infrastructure. The site sits within a region experiencing rapid natural gas pipeline expansion, creating potential interconnection opportunities for facilities requiring pipeline access.

Primary opportunity is natural gas interconnection via Gulf South Pipeline system or other regional gas transmission infrastructure. Proximity to Harvest CAM crude terminal demonstrates existing midstream infrastructure in immediate vicinity.

Recommended Next Actions

Phase 1: Infrastructure Mapping

- Obtain Louisiana DNR pipeline GIS data for 25-mile radius
- Request Gulf South Pipeline system maps from Boardwalk Pipeline Partners
- Contact Venture Global regarding Plaquemines LNG expansion plans and potential additional feed pipeline requirements

Phase 2: Interconnection Feasibility

- Engineering assessment of pipeline connection options
- Regulatory requirements evaluation (FERC/Louisiana PSC)
- Preliminary discussions with potential pipeline partners
- Cost estimates for connection infrastructure

Phase 3: Commercial Development

- Define specific use case for pipeline access
- Develop commercial framework with operators
- Pursue permits and approvals

REPORT CONTENTS

1. Regional Pipeline Infrastructure

1.1 Natural Gas Transmission Systems

Southeast Louisiana hosts one of the nation's densest natural gas pipeline networks, serving three distinct markets:

LNG Export Facilities - Cameron, Calcasieu Pass, and Plaquemines terminals requiring 10-15 Bcf/d of feed gas

Industrial Demand - Refineries, petrochemical plants, power generation (5-8 Bcf/d)

Interstate Commerce - Gas flows to Florida, Alabama, Mississippi markets

Major Interstate Systems:

- **Transco (Williams):** ~15 Bcf/d capacity, runs parallel to Mississippi River through East Bank parishes
- **Texas Eastern (Enbridge):** ~10 Bcf/d capacity, serves northeastern Louisiana and crosses into Mississippi
- **Tennessee Gas Pipeline:** ~11 Bcf/d capacity, reversed multiple segments southward to serve Louisiana LNG demand

Louisiana Intrastate Systems:

- **Gulf South Pipeline (Boardwalk):** Extensive Southeast Louisiana network serving local industrial demand and connecting to interstate systems
- **Louisiana Energy Gateway (LEG):** 1.8 Bcf/d system built by Williams (operational 2025) to serve Plaquemines LNG
- **NG3 Pipeline:** 2.2 Bcf/d system under development by Momentum Midstream (expected 2025)

1.2 Louisiana-Relevant Natural Gas Pipelines

Table 1: Recent Natural Gas Pipeline Projects

Pipeline	Operator	Capacity	Status	Purpose
Gulf Run	Enable Midstream	1.65 Bcf/d	Operational (2023)	Louisiana LNG feed
Louisiana Energy Gateway (LEG)	Williams	1.8 Bcf/d	Operational (2025)	Plaquemines LNG feed
NG3 Pipeline	Momentum Midstream	2.2 Bcf/d	Development (2025)	Louisiana LNG feed
TransCameron	Venture Global	1.9 Bcf/d	Under Construction	Cameron LNG feed
Gator Express	Venture Global	~2.0 Bcf/d	Operational	Plaquemines LNG header pipeline
East Lateral Xpress	Columbia Gulf (TC Energy)	0.725 Bcf/d	Operational (2025)	Ties into Gator Express
Hugh Brinson	Energy Transfer	TBD	Development	Louisiana LNG feed
Texas to Louisiana Energy Pathway	Williams	TBD	Development	Louisiana LNG feed expansion

Pipeline	Operator	Capacity	Status	Purpose
Gulf Xpress	TC Energy	Expansion	Operational (2019)	Louisiana market access

Key Observations:

- All recent pipeline construction in Louisiana is LNG-driven
- Plaquemines Parish has become a primary LNG export hub (Venture Global facilities)
- Williams, Energy Transfer, and Kinder Morgan dominate new infrastructure
- Capacity additions total 8+ Bcf/d in 2023-2026 period

1.3 Crude Oil Pipeline Systems

Louisiana crude oil pipeline infrastructure centers on the St. James Parish terminal complex, located approximately 60 miles upriver from Port Nickel. This hub receives Permian Basin crude via multiple long-haul systems and distributes to Gulf Coast refineries.

Table 2: Major Crude Oil Pipelines (Louisiana Terminal Points)

Pipeline	Operator	Capacity	Route	Status
Bayou Bridge	Energy Transfer	480 kbd	Nederland, TX to St. James	Operational (2019)
Capline (Reversed)	Plains All American	300 kbd	Patoka, IL to St. James	Operational (2022)
EPIC Pipeline	EPIC Midstream	900 kbd	Permian to Corpus Christi	Operational (2019)
Gray Oak	Enbridge	900 kbd	Permian to Corpus Christi/Ingleside	Operational (2020)
Wink to Webster	ExxonMobil	1,000 kbd	Permian to Houston area	Operational (2021)

Local Distribution and Terminal Systems:

- **Harvest CAM System:** Serves Belle Chasse Terminal (Harvest Midstream crude oil storage facility, formerly Phillips 66 Alliance Refinery) - operates within 5 miles of Port Nickel site
- **Colonial Pipeline Lines 1 & 2:** Major refined products corridor on East Bank (15-30 miles from site)
- **Collins Pipeline:** Refined products from Chalmette/Meraux refineries to Collins, MS (East Bank system, 15-20 miles)

Active Refineries (Upriver from Port Nickel):

- **St. Bernard Parish:** Chalmette Refining, Meraux (East Bank)
- **St. Charles Parish:** Norco Refinery, Valero St. Charles (East Bank)
- **St. John Parish:** Marathon Garyville (East Bank)
- **St. James Parish:** Gramercy/Convent area
- **East Baton Rouge Parish:** ExxonMobil Baton Rouge complex

Key Observations:

- St. James terminal complex is state's crude oil hub
- Multiple Permian takeaway systems deliver to Louisiana
- Port Nickel positioned downriver from refining complex
- Harvest CAM connects Belle Chasse crude terminal (no longer refining operations)

1.4 NGL and Petrochemical Pipelines

Natural Gas Liquids (NGLs) infrastructure in Louisiana primarily serves fractionation facilities and petrochemical plants along the Mississippi River chemical corridor.

Relevant NGL Systems:

- **Cayenne Pipeline:** Operates from Venice (Plaquemines Parish) north to chemical plants; crosses Mississippi River in Lower Plaquemines, estimated 10-20 miles from Port Nickel
- **ONEOK/EnLink Systems:** Serve St. James and Ascension Parish fractionators
- **Enterprise/Targa Systems:** Connect Gulf Coast NGL production to Louisiana processing facilities

2. Site-Proximate Infrastructure

2.1 Port Nickel Location Context

Geographic Position:

- **Parish:** Plaquemines (southernmost Mississippi River parish)
- **River Bank:** East Bank
- **River Mile:** Approximately 80 AHP (Above Head of Passes)
- **Distance to New Orleans:** ~20 miles downriver
- **Distance to Head of Passes:** ~75 miles upriver from mouth

Regional Context:

Port Nickel sits between two major energy infrastructure zones:

North (60-80 miles): New Orleans/Baton Rouge refining complex and St. James crude terminal hub

South (50-75 miles): Plaquemines LNG export terminal (Venture Global)

This positioning places the site along natural pipeline corridors connecting upstream production sources (Haynesville Shale, Permian Basin) to downstream markets (refineries, LNG terminals, chemical plants).

2.2 Pipelines Within 10-Mile Radius

Confirmed Proximate Infrastructure:

Harvest CAM Crude Oil System

- Distance: <5 miles (operates in vicinity of site)
- Operator: Harvest Midstream
- Product: Crude oil
- Terminus: Belle Chasse Terminal (crude oil storage, 1 million barrels capacity, two loading docks)
- Significance: Nearest confirmed major pipeline infrastructure to site

Gulf South Pipeline Natural Gas System (Estimated)

- Distance: 10-25 miles (extensive Louisiana network, exact route requires GIS verification)
- Operator: Boardwalk Pipeline Partners
- Product: Natural gas
- Significance: Major intrastate transmission system, potential primary interconnection candidate

Cayenne NGL Pipeline (East Bank)

- Distance: 10-20 miles to Mississippi River crossing point (estimated)
- Route: Venice to Toca (crosses river in lower Plaquemines)
- Product: Natural gas liquids
- Significance: Demonstrates pipeline infrastructure operates in Plaquemines Parish

2.3 Pipelines Within 25-50 Mile Radius

Major Systems (East Bank Corridor):

- **Transco (Williams):** 20-50 miles - Major interstate gas transmission, ~15 Bcf/d capacity
- **Colonial Pipeline Lines 1 & 2:** 15-30 miles - Refined products (gasoline, diesel, jet fuel)
- **Collins Pipeline:** 15-20 miles - Terminates at Chalmette/Meraux refineries, refined products to Collins, MS

Same Parish Infrastructure:

- **Gator Express Pipeline:** 50-75 miles south

- Operator: Venture Global
- Capacity: 2.9 Bcf/d
- Terminus: Plaquemines LNG export terminal
- Status: Operational
- **Critical Note:** Despite operating in same parish and serving major LNG facility, this pipeline has limited public documentation. Not found in RBN Energy's comprehensive 180-pipeline database. Suggests Venture Global built as dedicated facility pipeline rather than common carrier.

- **Louisiana Energy Gateway (LEG):** 50-75 miles south

- Operator: Williams
- Capacity: 1.8 Bcf/d
- Terminus: Plaquemines LNG
- Status: Operational 2025

2.4 Infrastructure Density Analysis

Pipeline Concentration Zones:

St. James Parish (60 miles northwest):

- Highest concentration crude oil infrastructure
- Multiple interstate crude systems terminate
- Refinery connections and distribution

East Bank Corridor (Port Nickel location and upriver):

- Major interstate gas transmission (Transco, Texas Eastern, Tennessee)
- Refined products pipelines (Colonial, Collins)
- Established pipeline right-of-ways
- Port Nickel positioned along this established corridor

West Bank (opposite side of river):

- Lower overall pipeline density than East Bank

- Some Gulf South gas network coverage

Mississippi River Pipeline Crossings:

Multiple pipelines cross the Mississippi River in Southeast Louisiana:

- Cayenne NGL (confirmed Plaquemines Parish crossing)
- Multiple gas transmission systems cross in Orleans/Jefferson parishes
- Crude systems cross upriver near St. James/Ascension
- Harvest CAM connects to Belle Chasse Terminal, demonstrating infrastructure in lower river area

Port Nickel's East Bank position provides direct access to the major pipeline corridor (Transco, Texas Eastern, Colonial, Collins) without requiring river crossings.

3. Planned and Development Projects

3.1 Natural Gas Pipeline Development

Louisiana leads the nation in new natural gas pipeline construction, driven entirely by LNG export expansion. Current and planned projects:

Table 3: Natural Gas Pipeline Development Projects

Pipeline	Operator	Capacity	Status	Purpose
TransCameron	Venture Global	1.9 Bcf/d	Under Construction	Cameron LNG expansion
NG3 Pipeline	Momentum Midstream	2.2 Bcf/d	Development	Louisiana LNG market
Hugh Brinson	Energy Transfer	TBD	Development	Louisiana LNG feed
Texas to Louisiana Energy Pathway Expansion	Williams	TBD	Development	Expand LEG capacity
Port Arthur-Louisiana Connector	Sempra	TBD	Development	TX/LA LNG facilities

3.2 LNG Export Context

Why So Many Pipelines?

Gulf Coast LNG export capacity is doubling in next 3 years:

- **Currently Operating:** 15.3 Bcf/d (Sabine Pass, Cameron, Calcasieu Pass, Corpus Christi, Freeport, Plaquemines)
- **Under Construction:** 12.3 Bcf/d additional (Golden Pass, Port Arthur, Rio Grande, expansions)
- **Total by 2028:** 27.6+ Bcf/d

Louisiana-Specific LNG Facilities:

Cameron LNG (Calcasieu Parish) - 12 MTPA (~1.6 Bcf/d) - Operational

Calcasieu Pass LNG (Calcasieu Parish) - 10 MTPA (~1.3 Bcf/d) - Operational

Plaquemines LNG (Plaquemines Parish) - ~2.9 Bcf/d - Operational/Under Construction - **Same parish as Port Nickel site**

This LNG buildout requires continuous natural gas supply, creating constant demand for pipeline infrastructure. Every new LNG facility requires dedicated feed pipelines capable of delivering 1-3 Bcf/d.

3.3 No Significant Crude or NGL Projects

Unlike natural gas (driven by LNG), crude oil and NGL pipeline construction in Louisiana has slowed:

- Permian-to-Gulf Coast crude infrastructure largely complete (2019-2021 buildout)
- NGL systems sufficient for current demand
- Focus has shifted to gas transmission

Implication: New pipeline opportunities in Louisiana are overwhelmingly natural gas-focused, specifically LNG-driven.

4. Regional Gas Flow Patterns

4.1 Supply Sources

Louisiana's natural gas supply comes from three primary sources:

Haynesville Shale Production (North Louisiana/East Texas)

- Production: 15+ Bcf/d
- Proximity: 150-250 miles north of Port Nickel
- Pipelines: Multiple systems connect Haynesville to Gulf Coast

Offshore Gulf of Mexico Production

- Production: 2-3 Bcf/d
- Pipelines: Connect to onshore Louisiana systems

Interstate Imports

- Gas from other regions flows through Louisiana to LNG facilities
- Systems like Transco, Texas Eastern bring gas from Appalachia and Midwest

4.2 Demand Centers

Louisiana natural gas flows to:

LNG Export (Primary Driver): 15+ Bcf/d and growing

Industrial/Petrochemical: 3-5 Bcf/d (refineries, chemical plants, power generation)

Interstate Export: 2-4 Bcf/d (to Southeast states)

Net Flow Pattern:

Gas flows **south and east** through Louisiana toward LNG terminals on Gulf Coast. This creates north-south pipeline corridors through Southeast Louisiana, with Port Nickel positioned along these flow paths.

5. Site Implications and Strategic Assessment

5.1 Why Pipeline Infrastructure Matters for Port Nickel

The site's position between New Orleans/Baton Rouge industrial complex (north) and Plaquemines LNG terminal (south) places it along established and growing energy infrastructure corridors.

Natural Gas Interconnection Opportunity:

- Gulf South Pipeline (10-25 miles) represents potential primary interconnection target
- System serves Louisiana intrastate market and connects to interstate pipelines
- Could provide natural gas supply for industrial facility or serve as interconnection point for gas-related operations

Crude Oil Context:

- Harvest CAM (<5 miles) demonstrates crude infrastructure in immediate vicinity
- System connects to Belle Chasse Terminal (crude oil storage and loading)
- St. James terminal complex (60 miles upriver) shows Louisiana's role in crude distribution

Strategic Positioning:

- East Bank location provides access to major pipeline corridor (Transco, Texas Eastern, Colonial)
- Same parish as Plaquemines LNG (largest growth driver in region)
- Positioned along established pipeline rights-of-way
- Direct access to East Bank infrastructure without river crossing requirements

5.2 Plaquemines LNG Feed Infrastructure

Gator Express Pipeline: Venture Global's Gator Express serves as the Plaquemines LNG header pipeline (main feeder system), with commissioning volumes around 2.0 Bcf/d. The pipeline operates in same parish as Port Nickel (approximately 50-75 miles south).

East Lateral Xpress: Columbia Gulf Transmission (TC Energy) recently placed into service the East Lateral Xpress project (725 MMcf/d capacity), which ties into Gator Express. FERC authorized partial service March 28, 2025, with flows beginning April 4, 2025.

Plaquemines LNG Feed Sources:

Plaquemines LNG receives natural gas from three main interstate connections:

TETCO (Texas Eastern Transmission)

Tennessee Gas Pipeline

Columbia Gulf Transmission (via East Lateral Xpress to Gator Express)

Site Relevance: As the closest major LNG facility to Port Nickel, Plaquemines LNG and its feed pipeline infrastructure represent the dominant natural gas demand driver in the immediate region. Understanding future expansion plans for Plaquemines LNG and potential additional feed capacity requirements is relevant to Port Nickel's gas infrastructure interconnection planning.

Recommended Action: Direct inquiry to Venture Global for information on Plaquemines LNG expansion timeline and potential additional feed pipeline requirements that could create regional interconnection opportunities.

5.3 Infrastructure Access Requirements

To evaluate Port Nickel's pipeline interconnection potential, need:

GIS Analysis: Precise pipeline locations within 25-mile radius

- Louisiana DNR pipeline GIS data
- Gulf South Pipeline route mapping
- Harvest CAM exact route location

Gulf South Pipeline Network Map:

- System operates extensively in Southeast Louisiana
- Specific route through Plaquemines Parish unknown
- Operator (Boardwalk) should be contacted for interconnection discussion

Plaquemines LNG Feed System Details:

- Gator Express route and capacity
- LEG (Williams) route
- Any planned expansions or additional feed pipelines

5.4 Recommended Next Steps

Phase 1: Infrastructure Mapping

- Obtain Louisiana DNR pipeline data overlays for site vicinity
- Request Gulf South Pipeline system maps from Boardwalk
- Identify Gator Express route through Venture Global inquiry
- Contract GIS analysis for precise distance measurements

Phase 2: Interconnection Feasibility

- Engineering assessment of pipeline connection options
- Regulatory requirements for interconnection (FERC if interstate, Louisiana PSC if intrastate)
- Preliminary discussions with potential pipeline partners (Boardwalk/Gulf South, Williams, others)
- Cost estimates for connection infrastructure

Phase 3: Commercial Development

- Define specific use case for pipeline access (gas supply, storage interconnection, etc.)
- Develop commercial framework with pipeline operators
- Pursue necessary permits and approvals

6. Conclusion

Southeast Louisiana's pipeline infrastructure is experiencing rapid expansion driven by LNG export growth. The Port Nickel site sits within this expanding corridor, positioned between the New Orleans/Baton Rouge industrial complex and Plaquemines Parish LNG terminals.

Key Takeaways:

Natural gas pipeline infrastructure dominates recent and planned development - all new construction is LNG-driven

Port Nickel's East Bank location positions site along the major Southeast Louisiana pipeline corridor with direct access to interstate gas transmission systems (Transco, Texas Eastern) and refined products pipelines (Colonial, Collins)

Proximate infrastructure includes Gulf South Pipeline (10-25 miles, estimated), Harvest CAM crude (<5 miles, confirmed), and Transco/Colonial systems (15-30 miles)

Critical information gap exists for Gator Express Pipeline - largest gas pipeline in Plaquemines Parish lacks complete public documentation

Pipeline interconnection feasibility requires GIS analysis and direct operator engagement to understand specific connection opportunities

Strategic Context:

Louisiana's LNG export infrastructure will continue expanding through 2028, requiring ongoing pipeline capacity additions. Port Nickel's location in this growth corridor creates potential opportunities for facilities requiring pipeline access, provided specific interconnection routes can be identified and engineered.

The primary limiting factor is lack of precise pipeline location data for the immediate Port Nickel vicinity. GIS analysis using Louisiana DNR data represents the critical next step for evaluating specific interconnection opportunities.

ANNEX A: SOURCES AND METHODOLOGY

Data Sources

Primary Sources

RBN Energy Midstream Infrastructure Database

- 180 pipeline inventory (natural gas, crude oil, NGLs)
- Louisiana-focused analysis: 24 pipelines with detailed specifications
- RBN analytics/maps database accessed December 2024 - January 2025
- Website: https://rbnenergy.com/analytics/maps

Industry Reports

- RBN Energy Deep Dive Reports (Marcellus/Utica, Permian-Gulf Coast, PADD Balances)
- LNG facility capacity data from operator investor presentations
- Market analysis reports (2024-2025)

Federal Sources

- FERC Major Pipeline Projects Database
- EIA Natural Gas Infrastructure Data
- U.S. Department of Energy LNG facility tracking

State Sources

- Louisiana Department of Natural Resources (DENR) Pipeline Program
- Louisiana State GIS Pipeline Feature Services (planned for future GIS analysis)

Operator Sources

- Pipeline operator websites and investor presentations
- Harvest Midstream operations data (Belle Chasse Terminal)
- Venture Global LNG facility data (Plaquemines LNG, Cameron LNG)
- Williams, Energy Transfer, Kinder Morgan project announcements

Secondary Sources

News and Industry Publications

- Oil & Gas Journal
- Pipeline & Gas Journal
- Houston Chronicle energy reporting
- BIC Magazine

Geographic Data

- Port Nickel GeoJSON parcel data (8 Louisiana parishes)
- Mississippi River mile marker system (AHP reference)
- Louisiana parish boundary data

Methodology

Pipeline Inventory Compilation

Initial Database Review: RBN Energy midstream infrastructure database (180 pipelines) filtered for Louisiana relevance

Geographic Filtering: Identified pipelines with Louisiana origins, destinations, or routes

Status Verification: Cross-referenced operational status, in-service dates, and capacity data

Categorization: Organized by commodity type (natural gas, crude oil, NGLs) and status (operational, under construction, development, cancelled)

Site Proximity Analysis

Location Establishment: Port Nickel coordinates (29.88°N, 89.93°W approximate), River Mile 80 AHP, East Bank

Distance Estimation: Calculated approximate distances to known pipeline systems using parish boundaries and operator route descriptions

Infrastructure Mapping: Identified confirmed proximate systems (Harvest CAM <5 miles, Gulf South 10-25 miles estimated, Transco/Colonial 15-30 miles)

Limitation Noted: Precise distances require GIS overlay analysis with Louisiana DNR pipeline data (recommended Phase 1 next step)

Market Analysis

LNG Capacity Assessment: Compiled operational and under-construction LNG facility data from operator sources

Pipeline Development Tracking: Identified all natural gas pipeline projects announced or under construction (2023-2027)

Flow Pattern Analysis: Assessed regional gas supply sources (Haynesville, offshore GOM, interstate) and demand centers (LNG, industrial, interstate export)

Limitations and Data Gaps

Known Limitations:

Gator Express Pipeline: Most significant gap - operates in Plaquemines Parish serving Plaquemines LNG but not found in RBN database; limited public documentation

Precise Pipeline Routes: Exact routes for Gulf South Pipeline and other intrastate systems in Plaquemines Parish not publicly mapped

Distance Measurements: Site proximity estimates based on parish geography and operator descriptions; GIS analysis required for precise measurements

Pipeline Capacity Utilization: Current utilization rates not available for all systems; capacity figures represent maximum design capacity

Data Confidence Levels:

• **High Confidence:** Major interstate pipeline systems (Transco, Texas Eastern, Tennessee), recently constructed LNG feed pipelines (LEG, Gulf Run, TransCameron), St. James crude terminal infrastructure

• **Moderate Confidence:** Distance estimates to Port Nickel, Gulf South Pipeline route approximations, crude oil system configurations in lower river area

• **Low Confidence (Requires Verification):** Gator Express route and specifications, precise Gulf South network mapping in Plaquemines Parish, exact Harvest CAM route near Port Nickel

Verification Approach

Cross-Reference Methodology

All pipeline data cross-referenced across multiple sources:

RBN Energy database (primary)

Operator websites and investor presentations (verification)

FERC filings where applicable (interstate systems)

News reports and industry publications (status updates)

Updates and Corrections

Refinery Status Correction:

• Initial draft incorrectly referenced Phillips 66 Alliance as active refinery

• Corrected: Phillips 66 Alliance converted to Belle Chasse Terminal (crude oil storage) in 2021, purchased by Harvest Midstream in 2023

• Source: Phillips 66 investor releases, Harvest Midstream announcements

Bank Location Verification:

• Initial draft incorrectly placed Port Nickel on West Bank

• Corrected: Port Nickel confirmed East Bank location

• Active refineries confirmed upriver from Port Nickel (Chalmette, Norco, Garyville, Baton Rouge, etc.)

ANNEX B: GLOSSARY OF TERMS

Bcf/d - Billion cubic feet per day (natural gas capacity measurement). Standard unit for large-scale gas transmission pipelines.

kbd - Thousand barrels per day (crude oil capacity measurement). One barrel = 42 U.S. gallons.

MMcf/d - Million cubic feet per day (natural gas capacity measurement). 1,000 MMcf/d = 1 Bcf/d.

MTPA - Million tonnes per annum (LNG capacity measurement). Approximately 0.13 Bcf/d per MTPA.

LNG - Liquefied Natural Gas. Natural gas cooled to approximately -260°F for volume reduction and ocean transport. One LNG cargo = ~3 Bcf gas equivalent.

NGL - Natural Gas Liquids. Hydrocarbon components extracted from natural gas: ethane, propane, butane, natural gasoline. Separated at processing plants and transported via dedicated NGL pipelines.

AHP - Above Head of Passes. Mississippi River mile marker system. Head of Passes is where river splits into delta channels. Port Nickel at Mile 80 AHP = 80 miles above river mouth.

Interstate Pipeline - Pipeline crossing state boundaries, regulated by Federal Energy Regulatory Commission (FERC). Requires FERC authorization for construction and interconnections.

Intrastate Pipeline - Pipeline operating within single state, regulated by state utility commission (Louisiana Public Service Commission). Generally less regulatory burden than interstate systems.

FERC - Federal Energy Regulatory Commission. Regulates interstate natural gas and oil pipelines, including construction authorization, rates, and interconnection.

Right-of-Way - Legal authorization to construct and operate pipeline across property. Established rights-of-way make new pipeline construction easier through shared corridors.

Interconnection - Physical connection between two pipeline systems allowing gas/crude transfer. Requires technical, commercial, and regulatory approval.

Haynesville Shale - Major natural gas producing region in North Louisiana and East Texas. One of nation's most productive gas fields, supplying Louisiana LNG facilities.

Permian Basin - Major oil and gas producing region in West Texas and New Mexico. Source of crude oil delivered to Louisiana via long-haul pipelines.

PADD 3 - Petroleum Administration for Defense District 3. Gulf Coast region including Louisiana, Texas, Arkansas, New Mexico, Alabama, Mississippi. Primary U.S. refining and petrochemical center.

LOOP - Louisiana Offshore Oil Port. Deepwater crude oil terminal in Gulf of Mexico offshore Louisiana. Can receive ultra-large crude carriers.

Y-Grade - Raw mixed NGL stream before fractionation into individual components. Transported from gas processing plants to fractionators.

Fractionator - Facility that separates mixed NGL stream into individual products (ethane, propane, butane, natural gasoline). Louisiana has multiple large fractionation facilities.

Common Carrier - Pipeline offering services to multiple shippers on non-discriminatory basis. Subject to regulatory oversight of rates and access.

Dedicated Facility Pipeline - Pipeline built to serve specific facility (e.g., LNG terminal), may not offer common carrier service. Gator Express appears to be this type.

ANNEX C: PIPELINE DATA TABLES

Table A-1: Louisiana-Relevant Natural Gas Pipelines (Comprehensive)

Pipeline Name	Operator	Capacity	Diameter	Status	ISD/Expected	Significance
Gulf Run	Enable Midstream	1.65 Bcf/d	42 in	Operational	Q4 2023	Louisiana LNG feed
Louisiana Energy Gateway (LEG)	Williams	1.8 Bcf/d	TBD	Operational	2025	Plaquemines LNG feed
NG3 Pipeline	Momentum Midstream	2.2 Bcf/d	TBD	Development	2025	Louisiana LNG market
TransCameron	Venture Global	1.9 Bcf/d	TBD	Under Construction	2025-2026	Cameron LNG feed
Gator Express	Venture Global	2.9 Bcf/d	TBD	Operational	Operational	Plaquemines LNG feed
Hugh Brinson	Energy Transfer	TBD	TBD	Development	2026+	Louisiana LNG feed
Texas to Louisiana Energy Pathway	Williams	TBD	TBD	Development	2025-2026	LEG expansion
Gulf Xpress	TC Energy	Expansion	TBD	Operational	2019	Louisiana market
Southwest Louisiana Supply Project	Tennessee Gas	Reversal	TBD	Operational	2018	Reversed to Louisiana
Trunkline Reversal	Energy Transfer	Reversal	TBD	Operational	2017	Reversed to Louisiana
Port Arthur-Louisiana Connector	Sempra	TBD	TBD	Development	2026+	TX/LA LNG connection
Gulf Coast Express	Kinder Morgan	2.0 Bcf/d	42 in	Operational	2019	Permian to TX coast

Table A-2: Louisiana Crude Oil Terminal and Distribution Systems

System Name	Operator	Function	Location	Status
Bayou Bridge	Energy Transfer	Long-haul crude	Nederland, TX to St. James	Operational
Capline (Reversed)	Plains All American	Long-haul crude	Patoka, IL to St. James	Operational
Harvest CAM	Harvest Midstream	Local crude distribution	Belle Chasse Terminal area	Operational
Colonial Lines 1 & 2	Colonial Pipeline	Refined products	Houston to East Coast	Operational
Collins Pipeline	Collins Pipeline	Refined products	Chalmette to Collins, MS	Operational

Table A-3: Louisiana LNG Facilities

Facility	Operator	Parish	Capacity (MTPA)	Capacity (Bcf/d)	Status
Cameron LNG	Sempra	Calcasieu	12	~1.6	Operational
Calcasieu Pass LNG	Venture Global	Calcasieu	10	~1.3	Operational
Plaquemines LNG	Venture Global	Plaquemines	~22	~2.9	Operational/Expanding

Table A-4: Active Refineries (Louisiana)

Refinery	Location	Parish	Bank	Approximate Capacity
Chalmette Refining	Chalmette	St. Bernard	East	190,000 BPD
Meraux Refinery	Meraux	St. Bernard	East	~135,000 BPD
Valero St. Charles	Norco	St. Charles	East	340,000 BPD
Norco Refinery	Norco	St. Charles	East	Component of above
Marathon Garyville	Garyville	St. John	East	578,000 BPD
ExxonMobil Baton Rouge	Baton Rouge	East Baton Rouge	East	520,000 BPD

ANNEX D: REFERENCES

Industry Sources

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